QUICK CARD · WIND TUNNEL (OTHER LAB EQUIPMENT)

Wind-tunnel fan

SINAMICS V20 1AC 230 V, 0.75 kW · 3~ fan motor in delta · § = section on the web page

Full page: **scan the QR code** or go to **ravo10.github.io/USN-kongsberg-lab-mech** → Equipment → Other.

NOT YET COMMISSIONED. Do not run the fan until the installation has been approved by the electrician and the lab supervisor.

SAFETY

- Keep hands, hair and loose items **away from the fan**
- **E-STOP** (red mushroom) cuts power to the drive
- Never above **50 Hz** (blades and bearings)
- After power-off: **wait 5 min** before opening the drive

A. Start §5

1. Tunnel and fan guard clear, nothing loose in the inlet.
2. Speed knob at **minimum**, RUN switch **off**.
3. E-stop released, press **START** (drive gets power, display on, no fault).
4. RUN switch **on**.
5. Turn the speed knob up **slowly**.

B. Stop §5

6. Turn the knob down, RUN switch **off** (fan ramps down in ~20 s).
7. Emergency: press **E-STOP**. After reset the fan does **not** restart until RUN is switched off and on again.
8. After use: E-stop in or mains off.

C. Drive settings §5

Speed knob	0–10 V = 0–50 Hz
Min / max	5 Hz / **50 Hz**
Ramp up / down	15 s / 20 s
Control	Quadratic V/f (fan macro AP020)
Show current	r0027 (A)
Full parameter list	workbook, sheet **V20 Wiring**

D. Faults §5

F85	Motor too hot (thermal contact). Let it cool, then press **RESET**.
Does not start	E-stop released? START pressed? RUN on? Fault on display?
Wrong direction	Stop. Electrician swaps two motor phases.
Other F-code	Note the code, switch off, tell the lab supervisor.

E. Motor leads (for the electrician) §3–4

V20 U	Black + Red
V20 V	Blue + Orange
V20 W	Brown + Grey
PE	Green/yellow
Thermal contact	2 × pale green → relay K1

F. Before the first run §6

- ☐ Insulation test 500 V passed
- ☐ Wiring approved by the electrician
- ☐ Rotation direction checked at ~10 Hz
- ☐ Thermal trip test: F85 appears, RESET works
- ☐ E-stop test: no restart until RUN off/on
- ☐ Current at 50 Hz noted (r0027) and motor data updated